

A Literature-Implied Negative Answer to the Cotype-Only Helly Question

Math discovery packet

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1 Status

This is a lightweight *literature-implied answer*, not an original solution. The source paper is

G. Ivanov, *No-dimensional Helly's theorem in uniformly convex Banach spaces*, arXiv:2409.05744.

In the Introduction, immediately before Section 2, the paper asks:

Is there a version of the no-dimensional Helly's theorem and a corresponding dual version of Maurey's lemma in a Banach space of cotype $q \geq 2$?

The supporting paper is

G. Ivanov, *On Banach Spaces with the Helly Approximation Property*, arXiv:2603.22743.

2 Identification

ArXiv:2603.22743 defines a Helly approximation sequence exactly in the “no-dimensional Helly” sense: every finite family of convex sets whose k -subfamilies meet the unit ball has a common point in the r_k -neighborhoods of all members. The space has the Helly approximation property when one can choose $r_k \rightarrow 0$.

The supporting paper proves the following two facts.

1. Theorem 1.1 gives a no-dimensional Helly theorem when X^* has Rademacher type $p > 1$, with

$$r_k(X) = 6T_p(X^*)k^{-1+1/p}.$$

2. Theorem 1.3 characterizes the property: X has the Helly approximation property if and only if X^* , equivalently X , has non-trivial Rademacher type.

Consequently, the correct qualitative hypothesis is non-trivial type, not cotype alone. Indeed, $L_1[0, 1]$ has cotype 2, while it has only trivial type because it contains ℓ_1^n 's uniformly, for example on disjoint measurable sets. By the characterization in arXiv:2603.22743, $L_1[0, 1]$ does not have the Helly approximation property. Therefore no cotype-only theorem can guarantee a no-dimensional Helly radius tending to zero.

3 Interpretation

This also explains the failure of the requested “dual Maurey’s lemma” under cotype alone. The dual reduction in arXiv:2603.22743 turns Helly approximation into an averaging statement in X^* . For $X = L_1[0, 1]$, the dual is $L_\infty[0, 1]$, which has no non-trivial Rademacher type, so Maurey’s sampling mechanism is unavailable.

4 Scope

The supporting theorem does not rule out cotype appearing in sharper quantitative statements once non-trivial type is also present. It rules out the source question as a cotype-only principle: finite cotype, even cotype 2, is insufficient.

5 References

- G. Ivanov, *No-dimensional Helly’s theorem in uniformly convex Banach spaces*, arXiv:2409.05744.
- G. Ivanov, *On Banach Spaces with the Helly Approximation Property*, arXiv:2603.22743.
- J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces II: Function Spaces*, for the standard type and cotype facts used for L_1 .